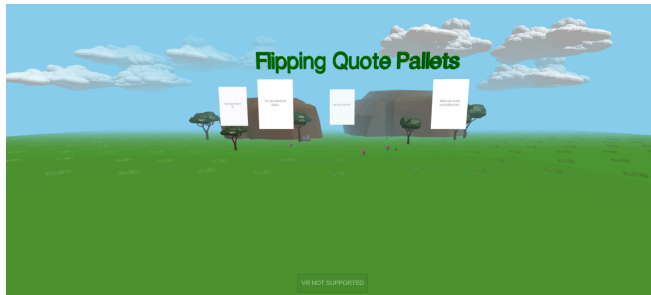


Assignment 3 – Flipping Quote Pallets

Objective:

The objective of this assignment was to develop an interactive VR-style 3D application using Three.js and WebXR concepts. The application demonstrates interactive flipping quote pallets placed inside an immersive 3D environment with animated natural elements and user interaction controls.

Visuals



Implementation:

The application was created using HTML, CSS, JavaScript, and the Three.js framework. For the 3D environment, multiple GLB models such as grass, trees, rocks, flowers, clouds, and butterflies were downloaded and placed inside the scene to create an immersive forest environment. The quote pallets were created using THREE.BoxGeometry. Dynamic canvas textures were used to display the quotes on one side and the author names on the reverse side of the pallets. Interactive functionality was implemented using raycasting. When a pallet was clicked, a smooth flip animation was applied on the Y-axis to reveal the reverse side. OrbitControls and WASD keyboard movement were also implemented for immersive scene navigation.

Additional animations such as moving clouds and realistic butterfly movement were added to improve the visual presentation and realism of the environment.

Flip Animation Logic:

Raycasting was used to detect mouse clicks on the quote pallets. When a pallet was clicked, its flip state was toggled and a smooth Y-axis rotation animation was applied using interpolation inside the animation loop.

```
cards.forEach ( (card) =>{  
  
    const targetRotation =  
  
        card.userData.flipped ? Math.PI: 0;  
    card.rotation.y +=(targetRotation - card.rotation.y) * 0.08;  
  
});
```

Text orientation was handled by creating a separate canvas for the front and back sides of each pallet. The front side displayed the quote while the reverse side displayed the author name.